

TRENTOS-M on NXP i.MX 8

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Demand by Maeringr

- 2x 10/100/1000 Ethernet
- RNG PRNG or TRNG or both ???

Maeringr is favouring the i.MX 8, especially the

- i.MX 8X (best match)
- i.MX 8MPlus (little less attractive due to higher power consumption)
- i.MX 8 (least attractive due to missing ECC)

Acceptable vendors

- Advantech
- Border Devices
- Congatec
- iWave
- Kontron
- MSC
- Toradex
- Variscite

Porting driver support

- Basic
 - UART
 - Ethernet
 - SD
 - TRNG
 - Secure boot code/firmware ???
- Future
 - SNVM
 - CAAM
 - ...

Platforms

In the following, all platform types from the i.MX 8 processor lines are highlighted, which adhere to the requirements mentioned by Maeringr.

Type	ARM Cores	Ethernet	Crypto	Source
i.MX 8	2x Cortex-A72 4x Cortex-A53 2x Cortex-M4F	2x 10/100 /1000	CAAM (RNG, using PRNG & TRNG)	https://www.nxp.com/products/processors-and-microcontrollers/arm-processors/i-mx-applications-processors/i-mx-8-processors/i-mx-8-family-arm-cortex-a53-cortex-a72-virtualization-vision-3d-graphics-4k-video:i.MX8

i.MX 8M Plus	4x Cortex-A53 1x Cortex-M7	2x 10/100 /1000 (1x TSN)	CAAM (PRNG ???)	https://www.nxp.com/products/processors-and-microcontrollers/arm-processors/i-mx-applications-processors/i-mx-8-processors/i-mx-8m-plus-arm-cortex-a53-machine-learning-vision-multimedia-and-industrial-iot:IMX8MPLUS
i.MX 8X	4x Cortex-A35 1x Cortex-M4F	2x 10/100 /1000	CAAM (RNG, using PRNG & TRNG)	https://www.nxp.com/products/processors-and-microcontrollers/arm-processors/i-mx-applications-processors/i-mx-8-processors/i-mx-8x-family-arm-cortex-a35-3d-graphics-4k-video-dsp-error-correcting-code-on-ddr:i.MX8X

- i.MX 8
 - "The purpose of the RNG is to generate cryptographically strong random data. It uses a true random number generator (TRNG) and a pseudo-random number generator (PRNG) to achieve true randomness and cryptographic strength. The RNG generates random numbers for secret keys, per message secrets, random challenges, and other similar quantities used in cryptographic algorithms." see IMX8QPAEC.pdf, p. 12
- i.MX 8M
 - "CAAM is a cryptographic accelerator and assurance module. CAAM implements several encryption and hashing functions, a run-time integrity checker, entropy source generator, and a Pseudo Random Number Generator (PRNG). The PRNG is certifiable by the Cryptographic Algorithm Validation Program (CAVP) of the National Institute of Standards and Technology (NIST)." see IMX8MDQLQIEC.pdf, p. 8

Vendors

Based on the requirements mentioned by Maeringr, the following platforms would apply.

CPU Type	Ethernet
i.MX 8	2x Gbit Ethernet AVB
i.MX 8M Plus	2x Gbit Ethernet AVB (one also supports TSN)
i.MX 8X	1-2x Gbit Ethernet AVB 1x 10/100 Ethernet

SBC

Vendor	SBC	Platform	CPU	RAM	Ethernet PHY	UART	SPI	SD /MMC	Source	Release Status	Company Location
Compulab	SBC-iMX8X	i.MX 8X	4x Cortex-A35 (@1.2 GHz) 1x Cortex-M4	up to 4GB	2x 10/100 /1000	1x UART-to-USB 1x RS-232 up to 2 UART 3.3 V	up to 4	up to 64GB eMMC	https://www.compulab.com/de/products/sbcs/sbc-imx8x-nxp-i-mx-8x-single-board-computer/	in production	Yokneam Illit (Israel)
Digi	Digi ConnectCore 8X SBC Pro	i.MX 8X (QuadXPlus)	4x Cortex-A35 (@1 GHz) 1x Cortex-M4	2GB	2x 10/100 /1000			16GB flash	https://www.digi.com/products/models/cc-sbp-wmx-jm8e	in production	Hopkins, MN (USA)
Seco	SBC-C57	i.MX 8X (DualXPlus /QuadXPlus)	up to 4x Cortex-A35 1x Cortex-M4F	up to 4GB	2x 10/100 /1000	1x RS-232 1x RS-485		up to 64GB QSPI NOR 1x eMMC 5.1 drive	https://www.seco.com/de/products/sbc-c57	in development	Arezzo (Italy)
Seco	SBC-C43	i.MX 8Quad /QuadPlus /QuadMax	up to 2x Cortex-A72 4x Cortex-A53 2x Cortex-M4F	up to 8GB	2x 10/100 /1000	1x RS-232 1x RS-485 1x TTL		up to 32GB eMMC 1x microSD slot	https://www.seco.com/de/products/sbc-c43	in development	Arezzo (Italy)

SoM

Vendor	SoM	Platform	CPU	RAM	Ethernet PHY	UART	SPI	SD /MMC	Source	Release Status	Company Location
Congatec	conga-SMX8 + - conga-SEVAL - conga-SMC1 /SMAR C- ARM	i.MX 8QuadPlus /QuadMax	up to 2x Cortex-A72 4x Cortex-A53 2x Cortex-M4F	up to 8GB	2x 10/100 /1000	4		eMMC 5.1 1x microSD	https://www.congatec.com/de/produkte/smarc/conga-smx8/	no information available	Deggendorf (Germany)
Congatec	conga-SMX8X + - conga-SEVAL - conga-SMC1 /SMAR C- ARM	i.MX 8X	up to 4x Cortex-A35 (@1.2 GHz) 1x Cortex-M4F (@264 MHz)	up to 4GB	2x 10/100 /1000	up to 4	1	eMMC 5.1 1x microSD	https://www.congatec.com/de/produkte/smarc/conga-smx8x/	no information available	Deggendorf (Germany)
FS Elektronik Systeme	efus MX8X + efus Basisboard	i.MX 8DualX /QuadXPlus	up to 4x Cortex-A35 (@1.2 GHz) 1x Cortex-M4F		2x 10/100 /1000	up to 4	up to 2	1x SD up to 64GB eMMC up to 512MB QSPI NAND	https://www.fs-net.de/de/produkte/efus/efusmx8x-mit-nxp-i-mx-8x-prozessor/	In production	Stuttgart (Germany)
Phytec	phyCORE-i.MX 8 (phyCORE-i.MX 8 Alpha Kit)	i.MX 8Quad i.MX 8QuadPlus i.MX 8QuadMax							https://www.phytec.de/produkt/system-on-modules/phycore-imx8/	Get a quote	Mainz (Germany)
Phytec	phyCORE-i.MX 8X + phyCORE-i.MX 8X Carrier Board	i.MX 8QuadXPlus i.MX 8DualXPlus i.MX 8DualX							https://www.phytec.de/produkt/system-on-modules/phycore-imx-8x/	tba	Mainz (Germany)
Toradex	Apalis iMX8 QuadPlus /QuadMax + Eagle Baseboard	i.MX 8QuadPlus i.MX 8QuadMax	up to 2x Cortex-A72 (@1.6 GHz) 4x Cortex-A53 (@1.26 GHz) 2x Cortex-M4F (@266 MHz)	up to 4GB (QuadMax) up to 2GB (QuadPlu s)		7	4		https://www.toradex.com/de/computer-on-modules/apalis-arm-family/nxp-imx-8 https://www.toradex.com/de/support/partner-network/hardware/23/eagle-baseboard-by-diamond-systems-corp	Limited stock & coming soon	Horw (Switzerland)
TQ Group	STKa8x (based on proprietary eval board + TQMa8x)	i.MX 8	2x Cortex-A72 (@1.6 GHz) 4x Cortex-A53 (@1.2 GHz) 2x Cortex-M4	up to 8GB	2x 10/100 /1000	up to 5	up to 4	up to 64GB eMMC	https://www.tq-group.com/de/produkte/tq-embedded/arm-architektur/stka8x/ https://www.tq-group.com/de/produkte/tq-embedded/arm-architektur/tqma8x/	Get a quote	Seefeld (Germany)
TQ Group	STKa8Xx (based on proprietary eval board + TQMa8Xx)	i.MX 8X	up to 4x Cortex-A35 (@1.2 GHz) 1x Cortex-M4	up to 2GB	2x 10/100 /1000	up to 4	up to 4	up to 64GB eMMC	https://www.tq-group.com/de/produkte/tq-embedded/arm-architektur/stka8xx/ https://www.tq-group.com/de/produkte/tq-embedded/arm-architektur/tqma8xx/ Angebot TQMa8Xx.UM.0005.pdf Additional comments see one row below	Available in 08/2020 699€ Starter Kit ~140€ for Module only	Seefeld (Germany)

TQ Group	STKa8XxS (based on MB-SMARC-2 + TQMa8XxS)	i.MX 8X	up to 4x Cortex-A35 (@1.2 GHz) 1x Cortex-M4	up to 2GB	2x 10/100 /1000	up to 2	1	up to 64GB eMMC	https://www.tq-group.com/de/produkte/tq-embedded/arm-architektur/stka8xxs/ Angebot TQMa8XxS.UM.0001.pdf ich interessiere mich für ihre embedded Module basierend auf i.MX8X - TQMa8XxS und TQMa8Xx. Können sie mir folgende Fragen beantworten: - Wie ist die Verfügbarkeit von diesen Boards und zugehörigen Carrier/Evaluation-Boards? Wir sind derzeit noch in der Redesign Phase welche im Juli dann abgeschlossen werden soll. Die Module und Starterkits sind dann somit ab August verfügbar. Geben sie mir einfach nur kurz Bescheid welche Cpu Variante sie verwenden wollen, dann kann ich ihnen das passende Starterkit anbieten. Die Starterkits haben einen einheitlichen Preis von 699€. - Welche Konfigurationen wären zuerst verfügbar? Wir werden mit Dual Plus und Quad Plus CPU's beginnen. Die Module sind dann per Default mit 1GB DDR3L (mit ECC) oder 1GB DDR4 (SMARC ohne ECC), 8GB eMMC ausgestattet. - Haben sie eine Roadmap für Varianten mit 2GB ECC Speicher? Wann wären diese Verfügbar? Diese Variante kann kundenspezifisch mit dem TQMa8Xx angeboten werden. Hinweis: der Speicherausbau mit 2GB und ECC ist auf Grund der Bauteile dann um einiges teurer! - Preisindikation für Module und Carrier für kleine Stückzahlen für die Entwicklung (<=10) Die Starterkits kosten für die Entwicklung 699€. In Serie (100St) kostet das Modul TQMa8Xx mit Quad Plus: 133€, mit Dual Plus: 129€ und das SMARC Modul mit Dual Plus: 145€ und mit Quadplus:149€ - Wir haben vor ein anderes Betriebssystem auf diese Plattformen zu portieren. Sind die Treiber und Treiberdokumentation als Open Source verfügbar (Linux Treiber wären ausreichend). Liefern sie ein angepasstes U-Boot mit? Alle Treiber die in unseren Linux BSP integriert sind, sind Open Source und können zum portieren in ihre OS verwendet werden.	Available in 08/2020 699€ Starter Kit ~150€ for Module only	Seefeld (Germany)
Variscite	VAR-SOM-MX8 + Symphony Board	i.MX 8	2x Cortex-A72 (@1.8 GHz) 4x Cortex-A53 (@1.2 GHz) 2x Cortex-M4F	up to 8GB	2x 10/100 /1000			up to 64GB eMMC	https://www.variscite.com/product/system-on-module-som/cortex-a72/var-som-mx8-nxp-freescale-i-mx8/ https://www.variscite.com/product/single-board-computers/symphony-board/	Starting from \$79 (get a quote)	Lod (Israel)
Variscite	VAR-SOM-MX8X + Symphony Board	i.MX 8X	4x Cortex-A35 (@1.2 GHz) 1x Cortex-M4F (@264 MHz)	up to 4GB	2x 10/100 /1000			up to 64GB eMMC	https://www.variscite.com/product/system-on-module-som/cortex-a35/var-som-mx8x-nxp-imx8x/ https://www.variscite.com/product/single-board-computers/symphony-board/	Early Access (Get a quote)	Lod (Israel)
Variscite	SPEAR-MX8 + VAR-SP8CustomBoard	i.MX 8	2x Cortex-A72 (@1.8 GHz) 4x Cortex-A53 (@1.2 GHz) 2x Cortex-M4F (@266 MHz)	up to 8GB	2x 10/100 /1000			up to 64GB eMMC	https://www.variscite.com/product/system-on-module-som/cortex-a72/spear-mx8-nxp-imx8/ https://www.variscite.com/product/single-board-computers/var-sp8customboard/	Early Access (Get a quote)	Lod (Israel)
Advantech	ROM-5620 + ROM-DB5901-SWA1	i.MX 8X	2 x Arm Cortex-A35 or 4 x Arm Cortex-A35	2GB LPDDR4	2x 10/100 /1000			16 GB eMMC NAND Flash for O.S. 8 MB QSPI NOR Flash for board information	https://www.advantech.eu/products/computer-on-module/rom-5620/mod_af8aa75b-54fb-4890-8ea6-a927fe0ea2e1 Distribution via https://www.systerra.de Angebot Systerra hensoldt_200701_44888 ROM-5620WU-OEA1E.pdf	333€ + 420€ available in storage Taiwan; approx. 3 weeks delivery time	Taiwan (Advantech) Systerra (Germany)

Existing sel4 support

The seL4 already provides support for i.MX 8 in a limited form and based on specific development boards.

SBC

According to <https://docs.sel4.systems/Hardware/>, seL4 currently supports the following NXP i.MX 8 platforms:

- MCIMX8M-EVKB
- IMX8MM-EVK

Board	CPU Name	CPU Type	Cores	CPU Clock	RAM	Ethernet	UART	SPI	SD /MMC	Source
MCIMX8M-EVKB	NXP i.MX 8MQuad	Cortex-A53 Cortex-M4	4 1	1.5 GHz 266 MHz	4GB	1x 10/100 /1000 Ethernet	1x (USB)	1x	1x microSD 1x eMMC	• https://www.nxp.com/design/development-boards/i-mx-evaluation-and-development-boards/evaluation-kit-for-the-i-mx-8m-applications-processor:MCIMX8M-EVK • we have one board in the office, used by Unknown User (chrhof01) for his master thesis
IMX8MM-EVK	NXP i.MX 8M Mini Quad	Cortex-A53 Cortex-M4	4 1	1.8 GHz 400 MHz	2GB	1x 10/100 /1000 Ethernet	1x (USB)	1x	1x	https://www.nxp.com/design/development-boards/i-mx-evaluation-and-development-boards/evaluation-kit-for-the-i-mx-8m-mini-applications-processor:8MMINILPD4-EVK

Resources

NXP provides a porting guide for

- Linux:
 - https://www.nxp.com/docs/en/user-guide/i.MX_BSP_Porting_Guide_Linux.pdf
 - https://www.nxp.com/docs/en/reference-manual/i.MX_Reference_Manual_Linux.pdf

Device drivers

Device drivers for the i.MX 8 exist in various other (OS) environments. In general, drivers could be ported over from

- Linux
- Genode GitHub
- U-Boot GitHub

The following device drivers are already included:

HW Type	HW Vendor	Data61 driver	Genode driver	U-Boot driver	Linux driver
Board			/repos/base-hw/lib/mk/spec/arm_v8/bootstrap-hw-imx8q_evk.mk /repos/base-hw/lib/mk/spec/arm_v8/core-hw-imx8q_evk.mk		
MMIO/IRQ			/repos/base/include/drivers/defs/imx8q_evk.h • GICv3 support		
GPIO					
Uart			/repos/base/include/drivers/uart/imx.h		
Timer					
SD-Card			/repos/os/src/drivers/sd_card/spec/imx8q_evk • based on the initial i.MX53 driver, but extended to i.MX 6, i.MX 7 and then i.MX 8		
USB			/repos/dde_linux/src/drivers/usb_host/spec/imx8q_evk • based on the existing i.MX driver		

NIC			/repos/dde_linux/fec.list /repos/dde_linux/recipes/src/fec_nic_drv /repos/dde_linux/src/drivers/nic/fec Update of fec network driver to version 4.16.3, which adds support for i.MX7 and i.MX8 SoCs		
Framebuffer			/repos/dde_linux/imx8_fb.list /repos/dde_linux/recipes/src/imx8_fb_drv /repos/dde_linux/src/drivers/framebuffer/imx8 Display Controller Subsystem (DCSS) & HDMI		

Genode supports the following i.MX 8 platforms:

Platform	SBC	Name	Source
i.MX 8	NXP MCIMX8X-EVK	imx8q_evk	https://www.nxp.com/design/development-boards/i-mx-evaluation-and-development-boards/evaluation-kit-for-the-i-mx-8m-applications-processor:MCIMX8M-EVK

Porting Proposal

An exemplary porting workflow to i.MX 8 could look like this.

- Acquire the board supported officially by the seL4: MCIMX8M-EVKB (<https://www.nxp.com/design/development-boards/i-mx-evaluation-and-development-boards/evaluation-kit-for-the-i-mx-8m-applications-processor:MCIMX8M-EVK>)
- Verify that existing seL4 support is working as expected & consider porting of missing parts required for TRENTOS-M
 - Re-use existing drivers for
 - UART
 - Timer
 - Interrupt
 - Port new drivers for
 - Ethernet
 - SD card
 - RNG
- As the MCIMX8M-EVKB is restricted to only 1x 10/100/1000 Ethernet, acquire a SBC/SoM that fits the requirements mentioned by Maering:
 - 2x 10/100/1000 Ethernet
 - RNG (TRNG & PRNG)
- Port TRENTOS-M to the acquired platform & adapt to missing points
 - Make use of 2nd Ethernet port